

AIMScribe v3 — Running It In The Cloud

What to rent, what it costs, and what has to change first · AIMS LAB · 4 October 2026

1 · The short answer

You are right that the server is not big enough. **A 100 GB disk holds three days of recording.** You need 2.16 TB for the study.

But the answer is **not** to rent a bigger disk. A 2.5 TB disk costs more every month than everything else put together. The answer is to stop keeping the audio on a disk at all, and keep it in **object storage** instead — the same kind of place you are already planning to put the FLAC copies.

Object storage costs roughly **one twentieth** of disk storage, never fills up, and needs no RAID. Once the audio lives there, the server only needs enough disk for the operating system, the databases and one consultation at a time — which is **about 100 GB**. The machine AIMS LAB is offering becomes exactly the right size.

That requires one change to the code, described in section 4. It is the single most valuable thing to build before this study starts, and everything cheap in this document depends on it.

If you do nothing	If you make that one change
Rent a server with a 2.5 TB disk. Roughly US\$350 a month , and the disk is a single point of failure you then have to mirror.	Keep a small server and put the audio in object storage. Roughly US\$35–110 a month depending on provider, with no disk to fill and no RAID to buy.

2 · How much audio there really is

Figure	Value
Consultations	18,000 over about 2.5 months
Clinic days	about 65 (six days a week)
Per day	about 277 consultations
Average file	120 MB
Audio per day	33 GB
All the WAV files	2,160 GB — 2.16 TB
All the FLAC copies (60% of WAV)	1,296 GB — 1.30 TB
Both together	3.46 TB
Both databases	about 30 GB, including backups

One thing worth checking before you buy anything. At 44.1 kHz, mono, 16-bit, the file size is not a guess — it is fixed arithmetic: **5.02 MB per minute**, always. So a 120 MB file is **24 minutes** of audio, and your 90–150 MB range is 18 to 30 minutes.

That is longer than the caps we sized for earlier — ten minutes at Aalo, fifteen to twenty at Amader Susastho, which averaged 11.4 minutes and 57 MB. At that average the study is **1.03 TB**, less than half. The difference is probably real: recording does not stop when the prescription is printed, it continues through the counselling, which is the part worth keeping. **This plan uses your 120 MB figure**, because planning short is the expensive mistake. If the true average turns out lower, every storage cost here halves.

Your other estimate — 8 or 10 MB per minute — would only be true at a higher sample rate or in stereo. At the format the research needs, it is 5.02 and cannot vary.

3 · Why 100 GB is not enough

The system keeps every finished recording on the server's own disk, forever. Nothing removes it. That is correct for a server with a large disk array, which is what it was designed for — and it is exactly what fails here.

Day	Audio on the disk	What happens
1	33 GB	fine
2	66 GB	fine
3	99 GB	the disk is full
4	—	the archive worker refuses to write and stops. Recordings stay on the doctors' laptops, which hold about 13 hours each, then those fill too.

The system fails **safely** — it pauses rather than losing anything, because a laptop only deletes its copy when the server proves it has a good one. But after about a day and a half of that, the laptops fill as well and recording stops in the clinics.

4 · The one change that fixes it

Today the worker does this:

Step	What happens	Where
1	Download the pieces, check each one	temporary space
2	Join them into one WAV, check it again	the server's disk
3	Write the clinical JSON beside it	the server's disk
4	Make the FLAC copy, encrypt it, upload it	cloud cold storage
5	Tell the server, which lets the laptop delete its copy	—
—	the WAV stays on the disk forever	this is the problem

The change is to add one step: **after the copy is verified, upload the WAV and its JSON to object storage, check they arrived, then delete them from the disk.**

Why this is small work, not a rewrite. The worker already uploads to object storage — that is how the FLAC copy gets to Amazon today, with its own request signing, its own verification, and a restore tool that fetches files back. The new step reuses all of it. What is needed is the upload-then-delete sequence, a column recording that it happened, and making the dashboard and the restore tool look in the new place. Realistically a few days of work, and it should be done and tested before the study starts, not during it.

The order matters and must not be shortened: upload, verify, *then* delete. Never delete on the strength of an upload that was merely accepted. This is the same rule the laptops already follow, and it is the rule that has kept every recording so far.

After the change, the server's disk holds only: the operating system (~20 GB), both databases and their backups (~30 GB), and one consultation being joined and compressed (~0.5 GB). **Under 60 GB, for ever.** A 100 GB disk is then comfortable.

5 · What the server actually has to do

Measured on our own bench, under a test twice as heavy as the real clinics (28 rooms recording at once):

Part	Memory in normal use	Ceiling we set
The server (API)	320 MB	1 GB
PostgreSQL (two databases)	240 MB	1 GB
Archive worker	250 MB	768 MB
Redis	23 MB	256 MB
Gateway (HTTPS)	40 MB	256 MB
Everything together	about 1 GB	3.4 GB

This is not a heavy system. All fourteen rooms together produce about 1.2 requests a second. The real work is compressing audio to FLAC, which uses processor rather than memory: roughly 2–3 minutes of one core per consultation, so about 11 hours of core time a day spread across 24 hours. Two cores handle it comfortably; four leaves room.

	Minimum that works	Recommended
Processor	4 cores	8 cores
Memory	8 GB	16 GB
Disk	100 GB SSD	160 GB SSD
Network	any modern link	—

The 8-core, 8 GB, 100 GB machine AIMS LAB is offering is enough — once the audio goes to object storage. That is worth saying plainly, because it may mean you do not need to rent a server at all.

6 · The three places your data lives

Place	What is in it	Why	Size
Transit store	Audio pieces on their way from the laptops	Deleted once the recording is archived and copied	small, self-clearing
Hot archive	The finished WAV, its clinical JSON, its manifest	This is the working dataset. It must be readable in seconds	2.16 TB
Cold copy	The lossless FLAC, encrypted before it leaves	Insurance. Rarely read, cheap to keep	1.30 TB

Do not put the hot archive in Amazon's deep archive tier. It is wonderfully cheap — about US\$1 per TB per month, as you found — but getting a file back takes **12 to 48 hours** and costs extra each time. That is right for insurance and wrong for the files your researchers open every day. The deep tier is for the FLAC copy only.

7 · The options, with prices

Prices are approximate, in US dollars, as of October 2026, and should be confirmed on the provider's own page before you commit — they change, and they differ by region. **Storage costs grow as the study fills it**, so the monthly figures shown are at the end of the study; the average over the 2.5 months is roughly half.

Option A — Keep the AIMS LAB server, rent only storage

Item	Choice	Monthly
Server	The 8-core / 8 GB / 100 GB machine you already have	\$0
Hot archive	Cloudflare R2, 2.16 TB	~\$32
Cold copy	Amazon S3 Glacier Deep Archive, 1.30 TB	~\$2
Getting data out	R2 charges nothing for downloads	\$0
Total at the end of the study		~\$34
Whole 2.5-month study		~\$50

Cheapest by a wide margin, and it uses hardware you already have. The trade-off is that the machine sits on UIU's network, so it needs the firewall and uptime arrangements from the separate CITS requirements document.

Option B — DigitalOcean, everything in one place

Item	Choice	Monthly
Server	Droplet, 4 vCPU / 8 GB / 160 GB	~\$48
Hot archive	Spaces object storage, 2.16 TB	~\$43
Cold copy	Amazon Glacier Deep Archive, 1.30 TB	~\$2
Backups	Droplet snapshots	~\$10
Total at the end of the study		~\$103
Whole 2.5-month study		~\$210

The simplest to operate: one provider, one bill, a clear control panel, and a Bangalore or Singapore region close to Dhaka. This is the option to choose if nobody wants to think about infrastructure.

Option C — Hetzner, cheapest rented server

Item	Choice	Monthly
Server	CCX23, 4 dedicated cores / 16 GB / 160 GB	~\$30
Hot archive	Hetzner Object Storage, 2.16 TB	~\$12
Cold copy	Amazon Glacier Deep Archive, 1.30 TB	~\$2
Traffic	20 TB included	\$0
Total at the end of the study		~\$44
Whole 2.5-month study		~\$95

The catch is distance. Hetzner is in Germany and Finland; a packet from Dhaka takes about 150 milliseconds each way. That is fine for uploading audio in the background and fine for the dashboard, but every clinic upload crosses half the world. It works — it is not where I would put a clinical system if a nearer option costs \$10 more.

Option D — Amazon, everything in AWS

Item	Choice	Monthly
Server	EC2 t3.xlarge, 4 vCPU / 16 GB (Mumbai)	~\$121

Disk	100 GB gp3	~\$8
Hot archive	S3 Standard, 2.16 TB	~\$54
Cold copy	S3 Glacier Deep Archive, 1.30 TB	~\$2
Total at the end of the study		~\$185
Whole 2.5-month study		~\$370

The most expensive, and the most capable. Worth it if you want managed databases, automatic scaling and one vendor for everything, or if UIU already has AWS credits. Mumbai (ap-south-1) is the nearest region. A one-year commitment cuts the server cost by about 40%.

Option E — Rent a big disk and change nothing

Item	Choice	Monthly
Server	Droplet, 4 vCPU / 8 GB	~\$48
Disk	2.5 TB block volume at \$0.10/GB	~\$250
Cold copy	Glacier Deep Archive	~\$2
Total at the end of the study		~\$300

Shown so the comparison is honest. It needs no code change, and that is its only advantage. It costs nine times Option A, the disk is a single copy you then have to protect, and it has to be sized correctly up front — if consultations run longer than expected, you are resizing a volume in the middle of a study.

8 · What I would do

Build the offload change, then take Option A. You already own a machine that is the right size once the audio moves off its disk, and the whole study then costs about fifty dollars of storage.

If the UIU machine is not available, or the network rules cannot be agreed, take Option B (DigitalOcean). It is \$210 for the study, needs no negotiation with anyone, and you can have it running in an afternoon. The extra cost buys you independence from the university's network policy, which may be worth more than the money.

Use Cloudflare R2 for the hot archive whichever server you pick. It is the cheapest per terabyte and, unusually, charges nothing to read data back. That matters here: the system reads every piece back to verify it, and your researchers will read the whole dataset repeatedly. On a provider that charges for downloads, that is a bill that grows with how much you use your own data.

9 · The costs nobody quotes

Trap	What it means	What to do
Download charges	Most providers charge \$0.08–0.12 per GB to read your own data out. Reading the 2.16 TB dataset once could cost \$200	Use R2 (free) or keep compute and storage with the same provider in the same region
Deep archive minimum	Amazon charges a minimum of 180 days for anything in Deep Archive. Delete a file after a month and you still pay six	Only put files there you intend to keep. Fine for your FLAC copies
Getting it back	Deep Archive retrieval takes 12–48 hours and costs about \$0.02 per GB. Restoring everything is ~\$26 and two days	Keep the hot archive for working, the cold copy for disaster only
Snapshots and backups	Usually billed separately, often forgotten until the first bill	Budget ~10% on top
The study is 2.5 months, the data is not	Storage keeps costing after the recording stops	Decide in advance how long the WAV stays hot. See below

What it costs after the study. Once collection ends you have a choice. Keep everything hot and readable: about \$34 a month on Option A. Or, when the research work is finished, delete the hot WAV files and keep only the FLAC copies in Deep Archive — **about \$2 a month, or \$24 a year**, for a lossless copy of every consultation. The FLAC decodes back to exactly the original samples, so nothing is lost by doing this; it only becomes slow to read.

10 · Running it without interruption

You said you want this to run rigorously and without interruption. Here is what actually threatens that, in the order it is likely to bite.

Risk	How the system already handles it	What you must still do
The server is down for an hour	Laptops keep recording and hold about 13 hours each. Nothing is lost	Nothing. This is designed for
The server is down for a day	Laptops fill up and recording stops in the clinics	Alerting, so an outage is noticed in minutes rather than at the end of a shift
Storage fills	The worker refuses to write rather than corrupting anything	Watch the dashboard. With object storage this stops being a real risk
A recording is never described by CMED	It is erased after 24 hours — correct, but silent	Check the unconfirmed count every morning. It should be zero
The disk or the provider loses data	There is an encrypted lossless copy at a second provider	Test a restore on day one, then monthly. A copy nobody has read is not a backup
Someone deletes something by mistake	Nothing protects against this	Turn on object versioning, and a delete lock on the cold bucket
The database is lost	Nightly encrypted backups	Test restoring one. Same rule

One machine or two? For a 2.5-month study, one server is the right answer. Two servers double the cost and more than double the complexity, and the thing that actually keeps clinics recording through an outage is not a second server — it is the buffer on each laptop, which you already have. Spend the effort on alerting and on testing restores instead.

11 · Security, in plain terms

What	How
Who can reach the server	Only ports 443 and 80 open. Administration over SSH from named machines only
Audio in transit	HTTPS everywhere; pieces encrypted on the laptop before they are sent
Audio at rest	Encrypted in the buckets. The cold copy is encrypted by us first, with a key the storage provider never has
The databases	Reachable only from inside the machine. Two separate roles, and the recordings database holds no patient names at all
Keys	In one file that is never committed. Both code repositories are public
Who is accountable	Every read of clinical data is logged, append-only

A word about putting patient audio on a foreign cloud. This is a governance question, not a technical one, and it should be answered by UIU's ethics committee before the first recording, not after. The encryption makes it defensible — the provider holds files it cannot read — but the decision is not an engineer's to make.

12 · What to do, in order

	Step	Who	Before
1	Decide the retention period — how long audio is kept	UIU ethics	buying anything
2	Confirm the real average consultation length from the first week of recordings	AIMS LAB	committing to a storage size

3	Build and test the offload change	AIMS LAB	the study starts
4	Pick a provider and open the accounts	AIMS LAB	the study starts
5	Deploy, then run the 28-room load test against the real server	AIMS LAB	any patient is recorded
6	Record one real consultation end to end	AIMS LAB	the clinics open
7	Restore one recording from the cold copy	AIMS LAB	day one, then monthly
8	Set up alerting on storage, failed jobs and silent recorders	AIMS LAB	the clinics open

13 · One page to remember

Question	Answer
How much audio?	2.16 TB of WAV, 1.30 TB of FLAC copies
Is 100 GB enough?	Only for three days — unless the audio moves to object storage, and then it is plenty
What has to change?	One step in the archive worker: upload the finished WAV to object storage, verify it, then delete it locally
Is the AIMS LAB server big enough?	Yes, after that change. 8 cores, 8 GB and 100 GB is the right size
Cheapest way	Keep that server, rent only storage — about \$50 for the whole study
Simplest way	DigitalOcean, one provider for everything — about \$210 for the study
Where should the hot files go?	Cloudflare R2 — cheapest, and free to read back
Where should the FLAC go?	Amazon Glacier Deep Archive — about \$2 a month
What does it cost afterwards?	\$34 a month to keep everything readable, or \$2 a month for the lossless copies alone
Biggest hidden cost	Download charges. R2 has none; most others do
Biggest real risk	Not storage — it is nobody noticing an outage. Alerting matters more than hardware
The one thing not to skip	Restore a recording from the cold copy on day one

Prepared by AIMS LAB. Prices are indicative and should be confirmed with each provider before purchase.